CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 1: Real Numbers

Topic-wise short notes with solved examples

This chapter has two ideas: every composite number breaks into primes in exactly one way, and that fact is used to find HCF and LCM and to prove that numbers such as $\sqrt{2}$ are irrational.

1 Numbers at a glance

- **Natural numbers**: $1, 2, 3, \ldots$ **Whole numbers**: $0, 1, 2, 3, \ldots$ **Integers**: $\ldots, -2, -1, 0, 1, 2, \ldots$
- **Rational number**: a number of the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.
- **Irrational number**: a number that cannot be written as $\frac{p}{q}$, such as $\sqrt{2}$, $\sqrt{3}$, π.
- **Real numbers**: all rational and irrational numbers together.
- **Prime number**: a number greater than 1 with exactly two factors, 1 and itself: $2, 3, 5, 7, 11, \ldots$
- **Composite number**: a number greater than 1 that is not prime: $4, 6, 8, 9, 10, \ldots$
- **Co-prime numbers**: two numbers whose only common factor is 1, such as 8 and 15.

REMEMBER

1 is neither prime nor composite. 2 is the only even prime number.

2 The Fundamental Theorem of Arithmetic

- **Statement**: every composite number can be written as a product of primes, and this factorisation is unique, apart from the order in which the primes are written.
- In general, a composite number x is written as $x = p_1 \times p_2 \times \cdots \times p_n$ with the primes in ascending order, and equal primes collected as powers.
- To find the prime factorisation, keep dividing by the smallest prime that divides the number.

Example 2.1 Express 3825 as a product of its prime factors.

SOLUTION

$3825 = 3 \times 1275 = 3 \times 3 \times 425$

$425 = 5 \times 85 = 5 \times 5 \times 17$

$3825 = 3^2 \times 5^2 \times 17$

Example 2.2 Explain why $7 \times 11 \times 13 + 13$ is a composite number.

SOLUTION

Take 13 common: $7 \times 11 \times 13 + 13 = 13\,(7 \times 11 + 1) = 13 \times 78$

The number has factors other than 1 and itself (13 and 78).

So it is a composite number.

Example 2.3 Can the number 12^n, where n is a natural number, end with the digit 0?

SOLUTION

A number ending in 0 is divisible by 10, so its prime factorisation must contain both 2 and 5.

$12^n = (2^2 \times 3)^n = 2^{2n} \times 3^n$. The only primes are 2 and 3.

By the uniqueness of prime factorisation, 5 can never appear.

No, 12^n cannot end with the digit 0 for any natural number n.

REMEMBER

A number ends in 0 only if its prime factorisation has both 2 and 5. This idea settles every question of the type "can a^n end with 0?"

3 HCF and LCM by prime factorisation

- **HCF** (highest common factor) = product of the **smallest** power of each **common** prime factor.
- **LCM** (least common multiple) = product of the **greatest** power of **every** prime factor that appears.
- For any two positive integers a and b:

$$\mathrm{HCF}(a, b) \times \mathrm{LCM}(a, b) = a \times b$$

Example 3.1 Find the HCF and LCM of 72 and 120 by prime factorisation, and verify that HCF × LCM = product of the numbers.

SOLUTION

$72 = 2^3 \times 3^2$ and $120 = 2^3 \times 3 \times 5$

HCF $= 2^3 \times 3 = 24$ (smallest powers of the common primes 2 and 3)

LCM $= 2^3 \times 3^2 \times 5 = 360$ (greatest powers of 2, 3 and 5)

Check: $24 \times 360 = 8640$ and $72 \times 120 = 8640$

HCF $= 24$, LCM $= 360$, and the relation is verified.

Example 3.2 The HCF of 91 and 126 is 7. Find their LCM.

SOLUTION

$$\text{LCM} = \frac{a \times b}{\text{HCF}} = \frac{91 \times 126}{7}$$
$$= 13 \times 126$$

LCM $= 1638$

Example 3.3 Find the HCF and LCM of 12, 18 and 30.

SOLUTION

$12 = 2^2 \times 3,\ 18 = 2 \times 3^2,\ 30 = 2 \times 3 \times 5$

$\text{HCF} = 2 \times 3 = 6$

$\text{LCM} = 2^2 \times 3^2 \times 5 = 180$

HCF $= 6$ and LCM $= 180$

REMEMBER

The relation HCF $\times$ LCM = product works for **two** numbers only. In the last example $12 \times 18 \times 30 = 6480$, but HCF $\times$ LCM $= 6 \times 180 = 1080$.

4 Word problems on HCF and LCM

- Use **HCF** when the question asks for the greatest or largest size, length or number that divides the given quantities exactly.
- Use **LCM** when the question asks for the least or smallest quantity divisible by all, or when events repeat and you need the time they next happen together.

Example 4.1 Two ropes of lengths 84 m and 126 m are to be cut into pieces of equal length, with nothing left over. Find the greatest possible length of each piece.

SOLUTION

The length must divide both 84 and 126, and be the greatest such number, so we need the HCF.

$84 = 2^2 \times 3 \times 7$ and $126 = 2 \times 3^2 \times 7$

$\text{HCF} = 2 \times 3 \times 7 = 42$

Each piece is 42 m long.

Example 4.2 Three bells ring at intervals of 9, 12 and 15 minutes. If they ring together at 8:00 a.m., when will they next ring together?

SOLUTION

They ring together after a time that is a multiple of 9, 12 and 15; the first such time is the LCM.

$9 = 3^2, 12 = 2^2 \times 3, 15 = 3 \times 5$

LCM $= 2^2 \times 3^2 \times 5 = 180$ minutes $= 3$ hours

They next ring together at 11:00 a.m.

5 Irrational numbers and their proofs

- **Theorem**: if a prime number p divides a^2, then p divides a, where a is a positive integer.
- The proofs below use the method of **contradiction**: assume the opposite of what is to be proved and show that this leads to something impossible.

Example 5.1 Prove that $\sqrt{2}$ is irrational.

SOLUTION

Assume, to the contrary, that $\sqrt{2}$ is rational. Then $\sqrt{2} = \frac{a}{b}$, where a and b are co-prime integers and $b \neq 0$.

Squaring: $2 = \frac{a^2}{b^2}$, so $a^2 = 2b^2$. Hence 2 divides a^2, and so 2 divides a.

Let $a = 2c$. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$. Hence 2 divides b^2, and so 2 divides b.

So 2 is a common factor of a and b. This contradicts the fact that a and b are co-prime.

Our assumption is wrong. Hence $\sqrt{2}$ is irrational.

Example 5.2 Prove that $3 + 2\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational.

SOLUTION

Assume, to the contrary, that $3 + 2\sqrt{5}$ is rational. Then $3 + 2\sqrt{5} = \frac{a}{b}$, where a and b are integers and $b \neq 0$.

Rearranging: $\sqrt{5} = \frac{a - 3b}{2b}$

Since a and b are integers, $\frac{a - 3b}{2b}$ is rational, so $\sqrt{5}$ is rational.

This contradicts the fact that $\sqrt{5}$ is irrational.

Hence $3 + 2\sqrt{5}$ is irrational.

- The proof that $\sqrt{3}$, $\sqrt{5}$ or $\sqrt{p}$ (for any prime p) is irrational follows the same five steps as for $\sqrt{2}$, with 2 replaced by the prime.
- The sum or difference of a rational number and an irrational number is irrational.
- The product or quotient of a non-zero rational number and an irrational number is irrational.

REMEMBER

In every proof, write "where a and b are co-prime" in the first step. The contradiction depends on it, and marks are given for it.

6 Common mistakes

- Writing 1 as a prime factor. 1 is not a prime number.
- Taking the greatest powers for HCF, or the smallest powers for LCM. It is the other way round.
- Using HCF $\times$ LCM = product for three numbers.
- In irrationality proofs, forgetting to say what the contradiction is.
- Mixing up when to use HCF and LCM in word problems: "greatest that divides" means HCF; "least that is divisible by" or "together again" means LCM.

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